Lecture 9 – Fault Localization

AAA705: Software Testing and Quality Assurance

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2026 Fall

Recall

- Non-Testable Programs
  - No oracle, or an oracle as hard as the program itself
- Metamorphic Testing
  - A relation between two runs replaces the expected output – solvers, databases, compilers
- Differential Testing
  - Another implementation is the oracle – DNNs, Nezha, JIT on/off, N+1-version testing
- Property-based Testing
  - Properties plus generated inputs – QuickCheck, Hypothesis, **shrinking**

The Test Failed. Now What?

`mid(2, 1, 3)` should return 2. It returns 1.

```
1  int mid(int x, int y, int z) {
2    int m = z;
3    if (y < z) {
4      if (x < y)       m = y;
5      else if (x < z) m = y;  // bug: m = x
6    } else {
7      if (x > y)       m = y;
8      else if (x > z) m = x;
9    }
10   return m;
11 }
```

- The failing test says the program is **wrong**. It does not say **which line**. Here there are 11; in a real project, thousands.

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```

- The failing test says the program is **wrong**. It does not say **which line**. Here there are 11; in a real project, thousands.
- **Today**: narrow it down. First shrink the **input**, then rank the **lines**.

Fault Localization



- **Fault localization**: from a failing test to the **location** of the fault.

Fault Localization



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- **Fault localization**: from a failing test to the **location** of the fault.
- By hand: `print`, `assert`, a breakpoint.
- Today: **automatically**, from the input, from the bug report, from the tests.

Contents

1. Delta Debugging (DD)

2. Information Retrieval based Fault Localization (IRFL)

3. Spectrum-based Fault Localization (SBFL)

4. Mutation-based Fault Localization (MBFL)

5. Learning to Localize

Contents

1. Delta Debugging (DD)
    - Recursive Delta Debugging – `ddmin`
    - Delta Debugging on Changes
    - Hierarchical Delta Debugging
    - Probabilistic Delta Debugging (ProbDD)
    - Delta Debugging for Program Debloating
2. Information Retrieval based Fault Localization (IRFL)
3. Spectrum-based Fault Localization (SBFL)
4. Mutation-based Fault Localization (MBFL)
5. Learning to Localize

Motivation

Printing this page (896 lines of HTML) crashed Mozilla. Which part is the cause?[1]

```
<td align=left valign=top>
<SELECT NAME="op sys" MULTIPLE SIZE=7>
<OPTION VALUE="All">All <OPTION VALUE="Windows 3.1">Windows 3.1
<OPTION VALUE="Windows 95">Windows 95 <OPTION VALUE="Windows 98">Windows 98
... (30 more operating systems) ...
<OPTION VALUE="other">other</SELECT></td> <td align=left valign=top>
<SELECT NAME="priority" MULTIPLE SIZE=7>
<OPTION VALUE="--">-- <OPTION VALUE="P1">P1 <OPTION VALUE="P2">P2
... (880 more lines) ...
```

[1] **[Mozilla bug 24735]** "*Printing causes mozilla to crash*" (2000). Numbers from **[TSE'02]**.

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... (880 more lines) ...
```

57 runs, line by line: `<SELECT NAME="priority" MULTIPLE SIZE=7>`

26 more, char by char: `<SELECT>`

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Delta Debugging (DD)

Delete a part and run again. Still fails? Keep it deleted. Repeat.[2]

- The result: a **small** input that still fails.

[2] **[TSE'02]** A. Zeller, R. Hildebrandt. "*Simplifying and Isolating Failure-Inducing Input.*"

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- The result: a **small** input that still fails.
- Easy to **read** and to **report**. Two reports that shrink to the same input are the **same bug**.

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- The result: a **small** input that still fails.
- Easy to **read** and to **report**. Two reports that shrink to the same input are the **same bug**.
- **Shrinking** in property-based testing is the same idea.

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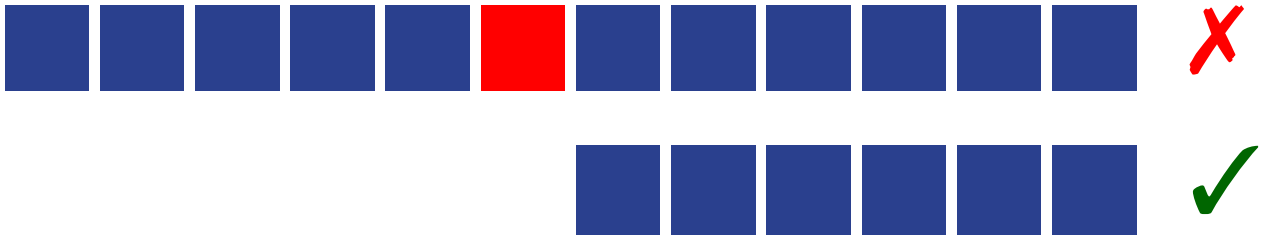
Delta Debugging

The simplest strategy is **binary search**. Red is the part that matters:



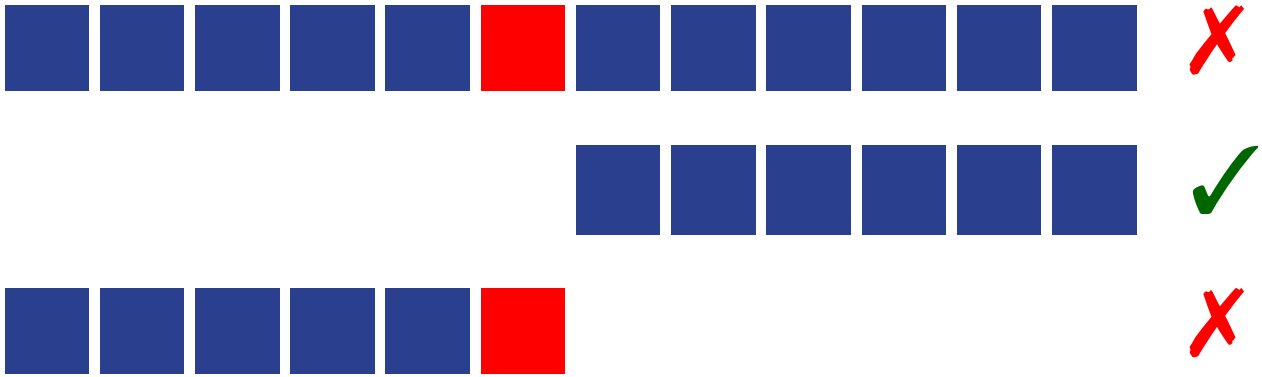
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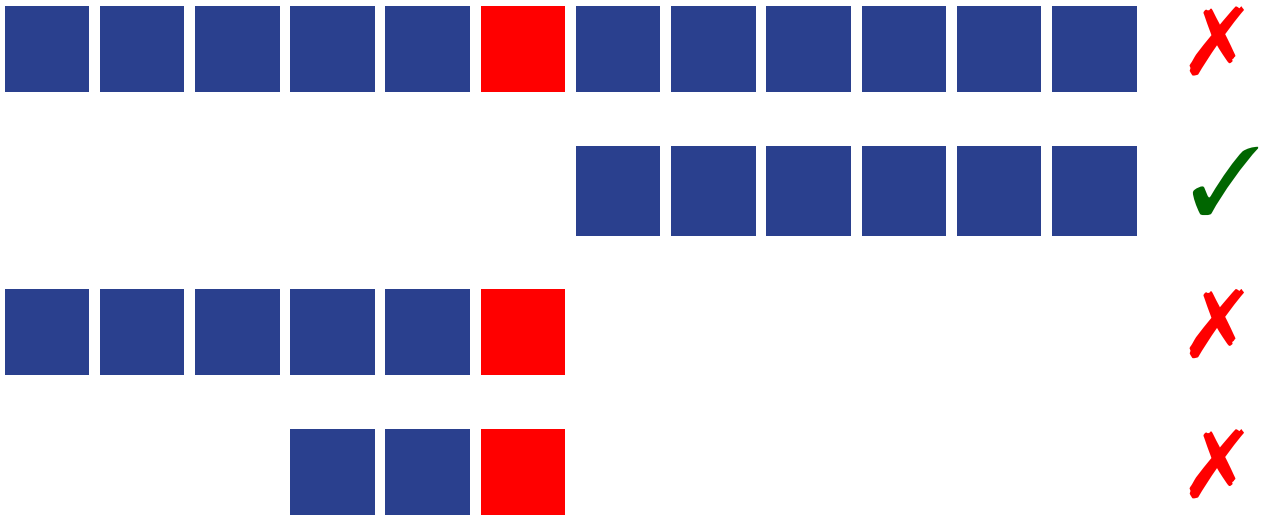
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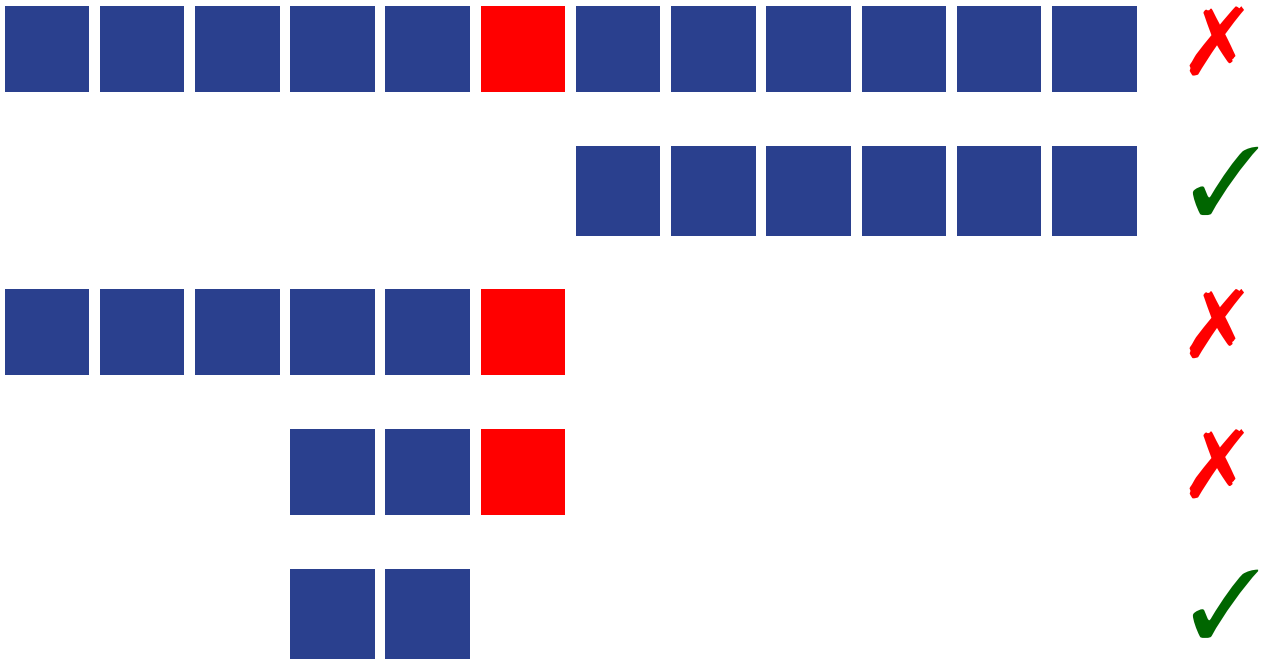
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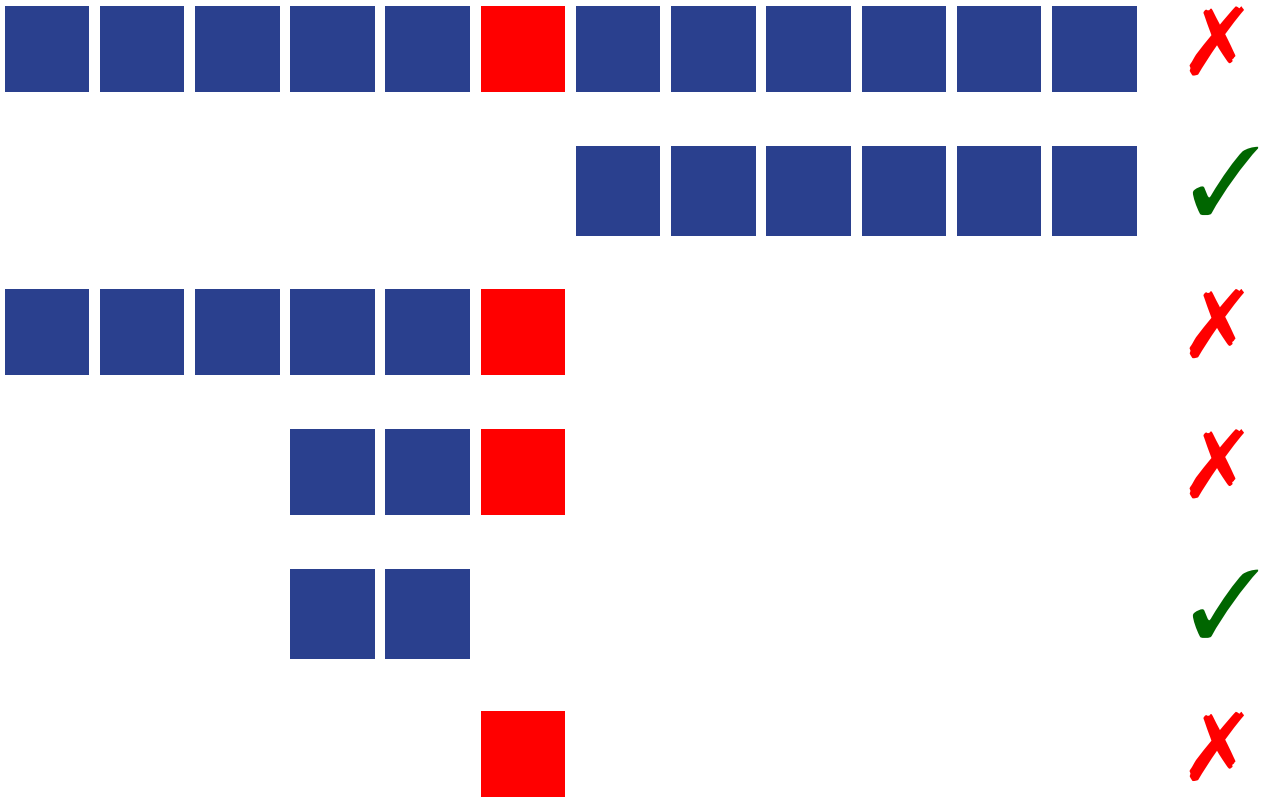
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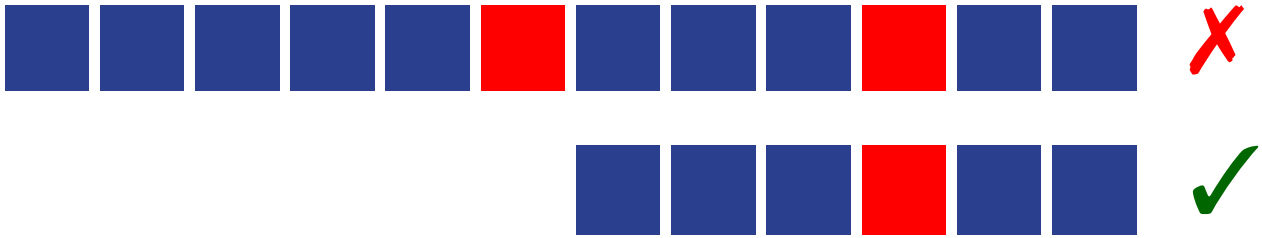
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But what if the failure needs **two** parts? Either half alone passes:



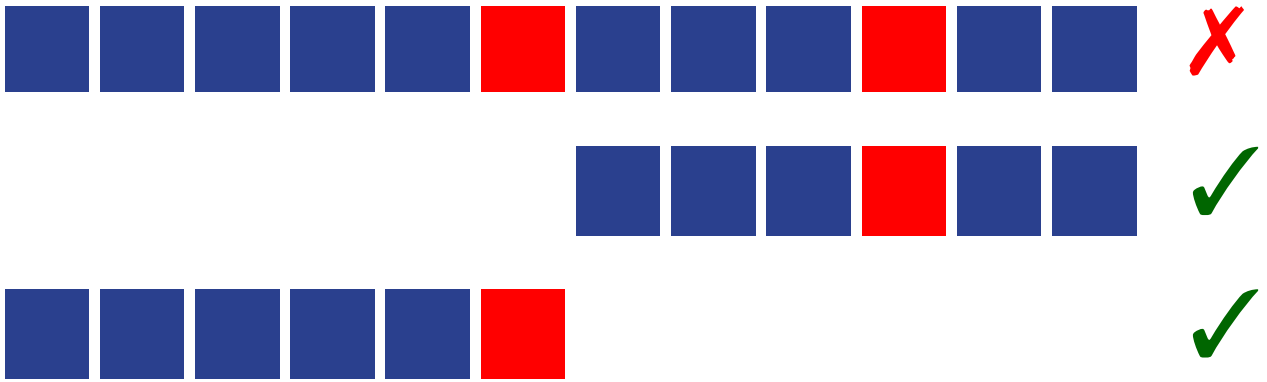
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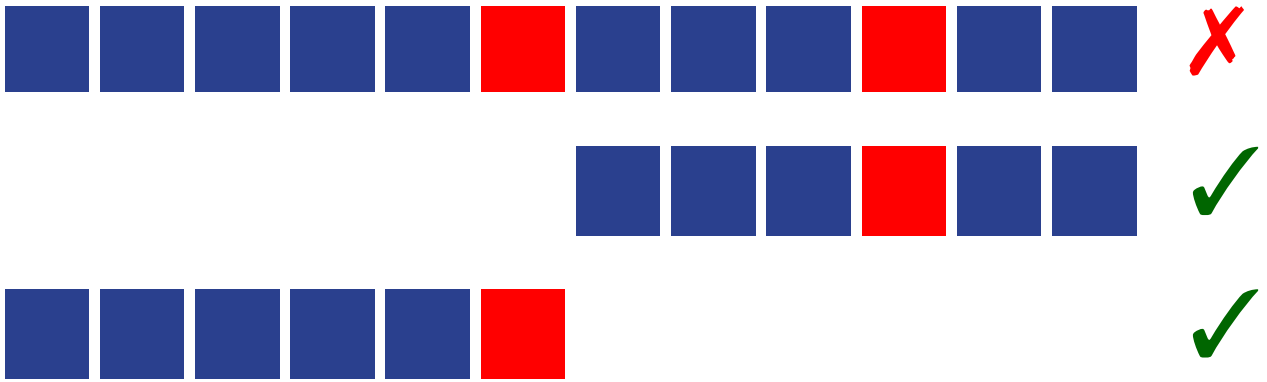
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So try to **keep** a chunk, then to **remove** a chunk. If neither works, cut **smaller** chunks.

Recursive Delta Debugging – ddmin

PLRG

Algorithm ddmin

Input: a failing test case t **Output:** a **1-minimal** failing test case t'

```
1:  function DDMIN(t)
2:      L ← the list of elements in t;    n ← 2
3:      while n ≤ |L| do
4:          ⟨c_1, ..., c_n⟩ ← split L into n chunks
5:          if ∃i ∈ [1, n]. test(c_i) = ✗ then
6:              ⟨L, n⟩ ← ⟨c_i, 2⟩                          ▷ keep a chunk
7:          else if ∃i ∈ [1, n]. test(L \ c_i) = ✗ then
8:              ⟨L, n⟩ ← ⟨L \ c_i, max(n − 1, 2)⟩          ▷ remove a chunk
9:          else if n < |L| then
10:             n ← min(2n, |L|)                           ▷ smaller chunks
11:         else
12:             break
13:     return the test case t' that corresponds to L
```

1-minimal: no **single** element can go. Not the smallest. $O(|t|^2)$ runs.

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1: **function** DDMIN(t)
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3: **while** $n \leq |L|$ **do**
4: $\langle c_1, \ldots, c_n \rangle \leftarrow$ split L into n chunks
5: **if** $\exists i \in [1, n].\ test(c_i) = \text{✗}$ **then**
6: $\langle L, n \rangle \leftarrow \langle c_i, 2 \rangle$ ▷ keep a chunk
7: **else if** $\exists i \in [1, n].\ test(L \setminus c_i) = \text{✗}$ **then**
8: $\langle L, n \rangle \leftarrow \langle L \setminus c_i, \max(n-1, 2) \rangle$ ▷ remove a chunk
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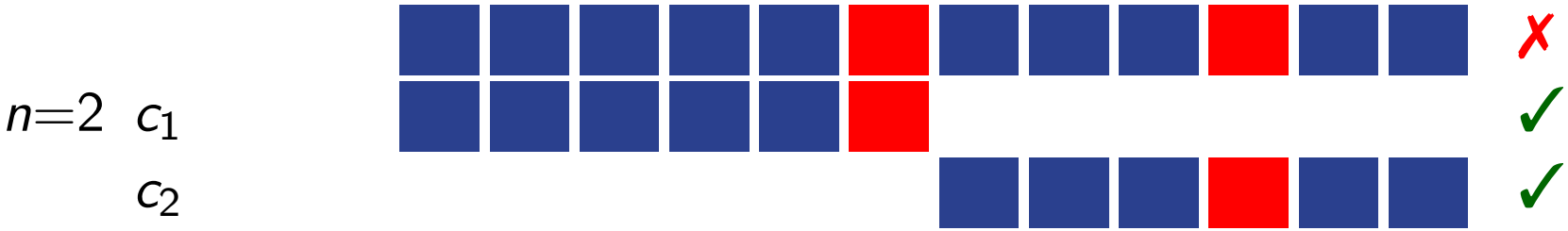
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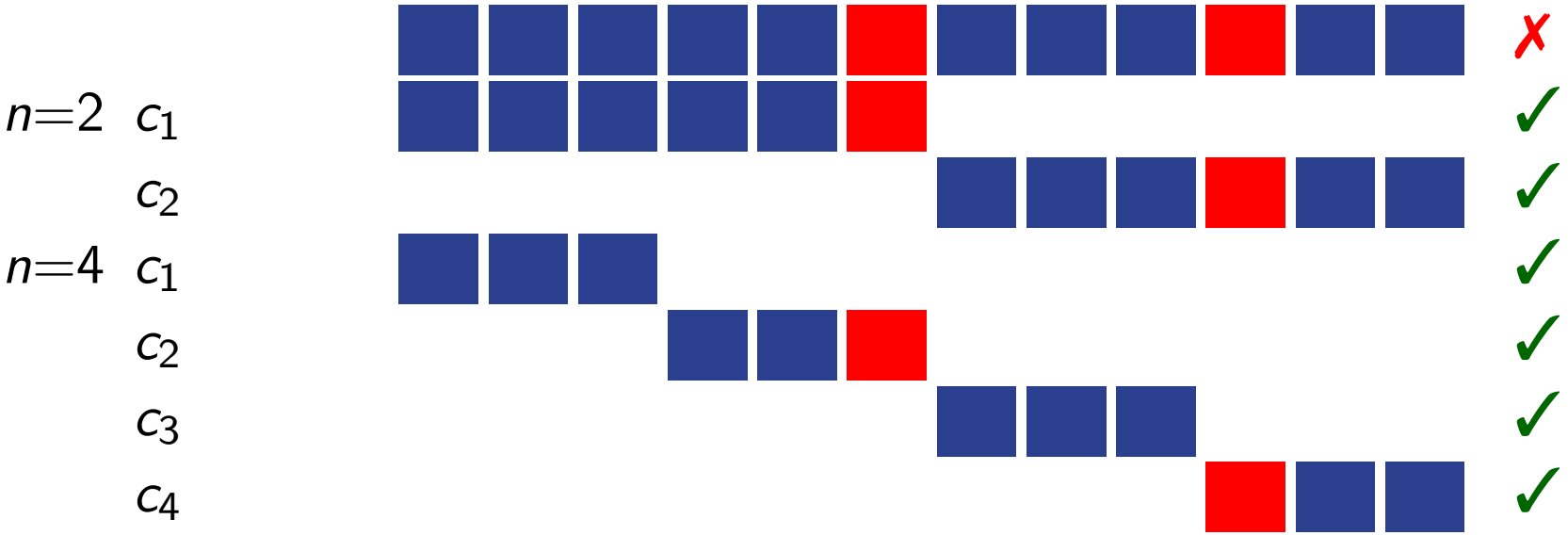
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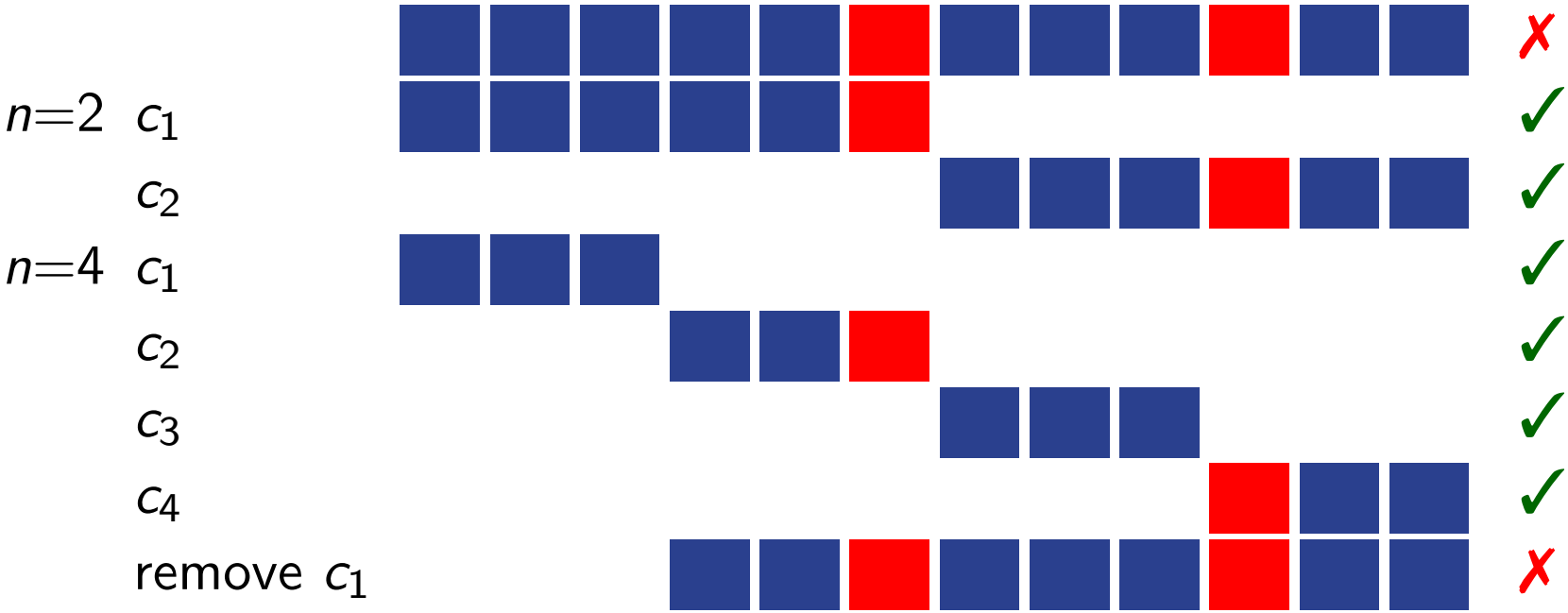
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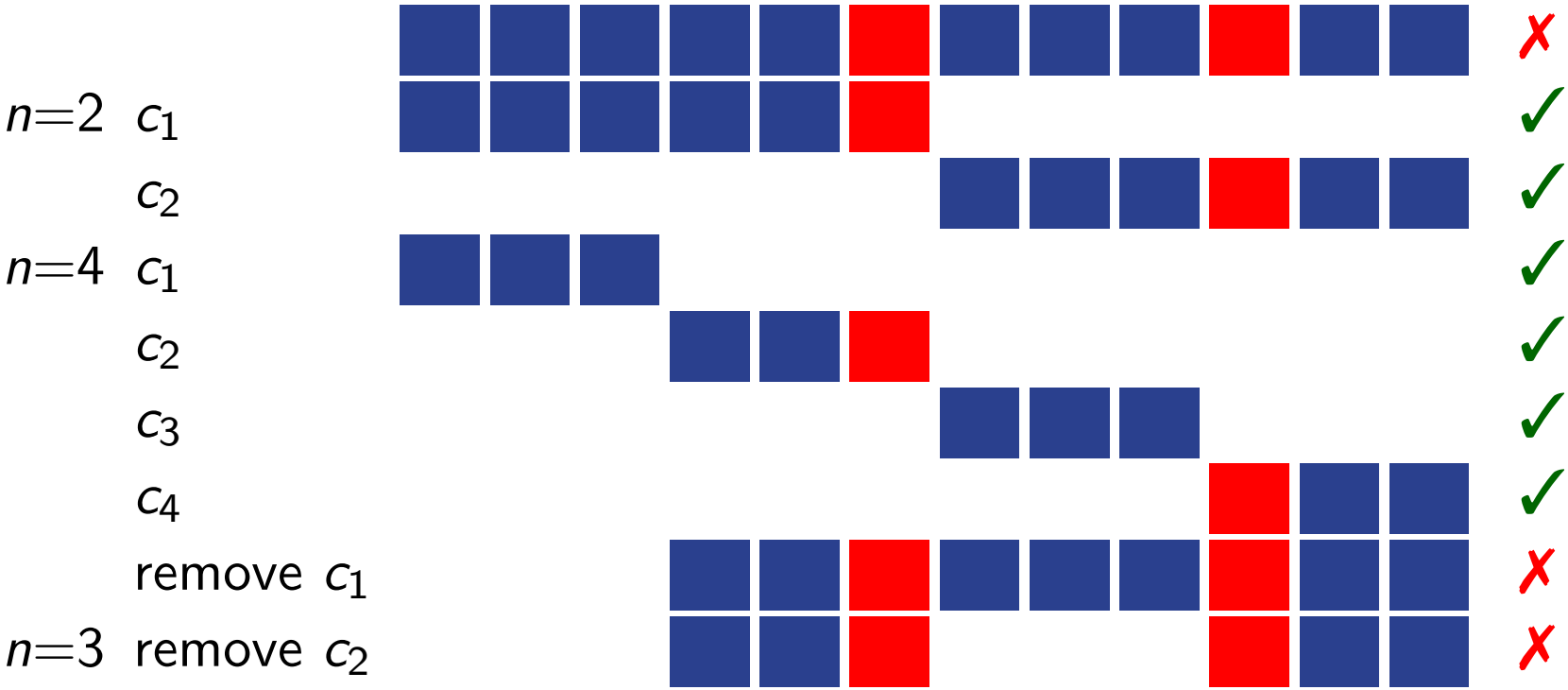
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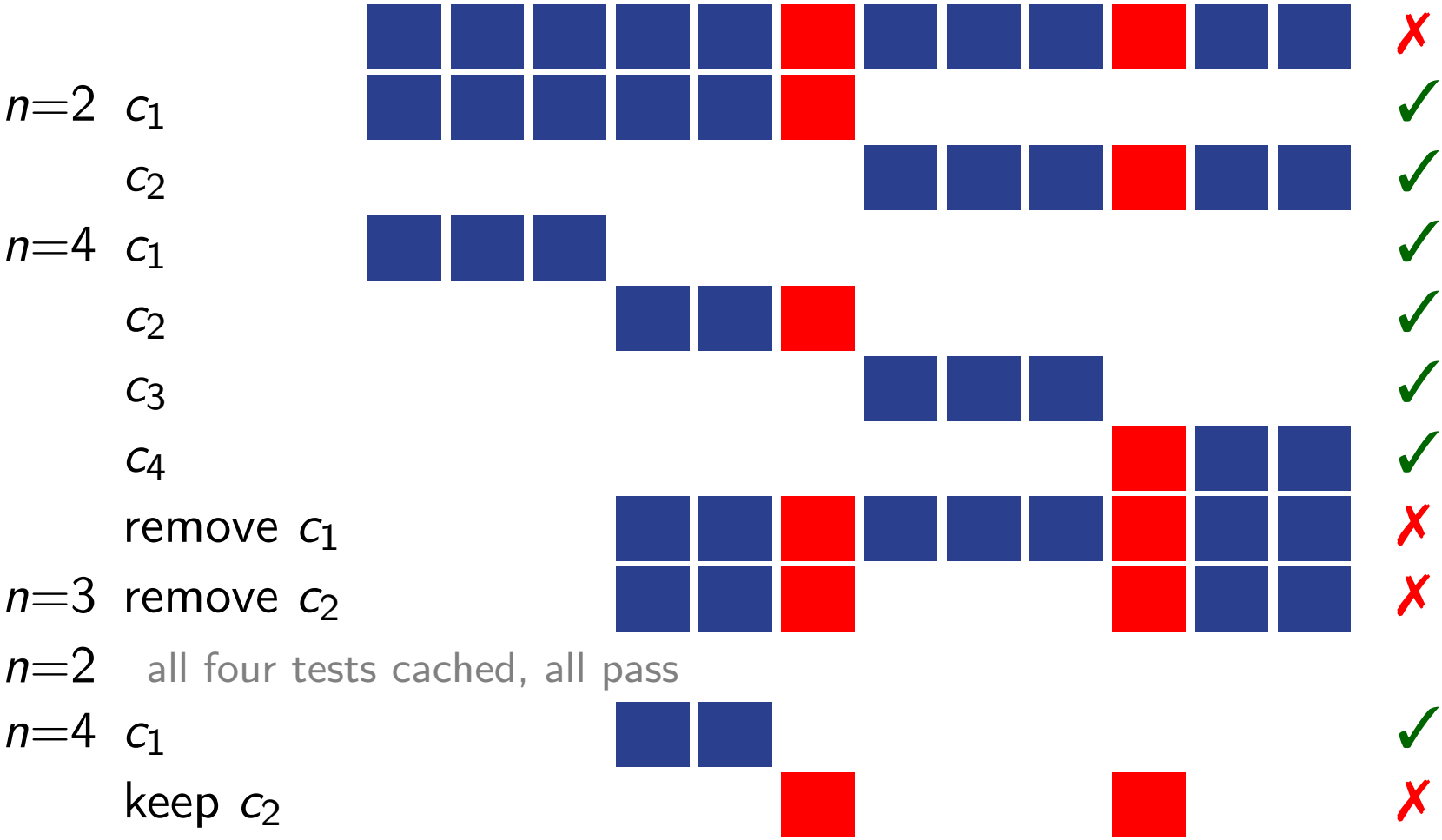
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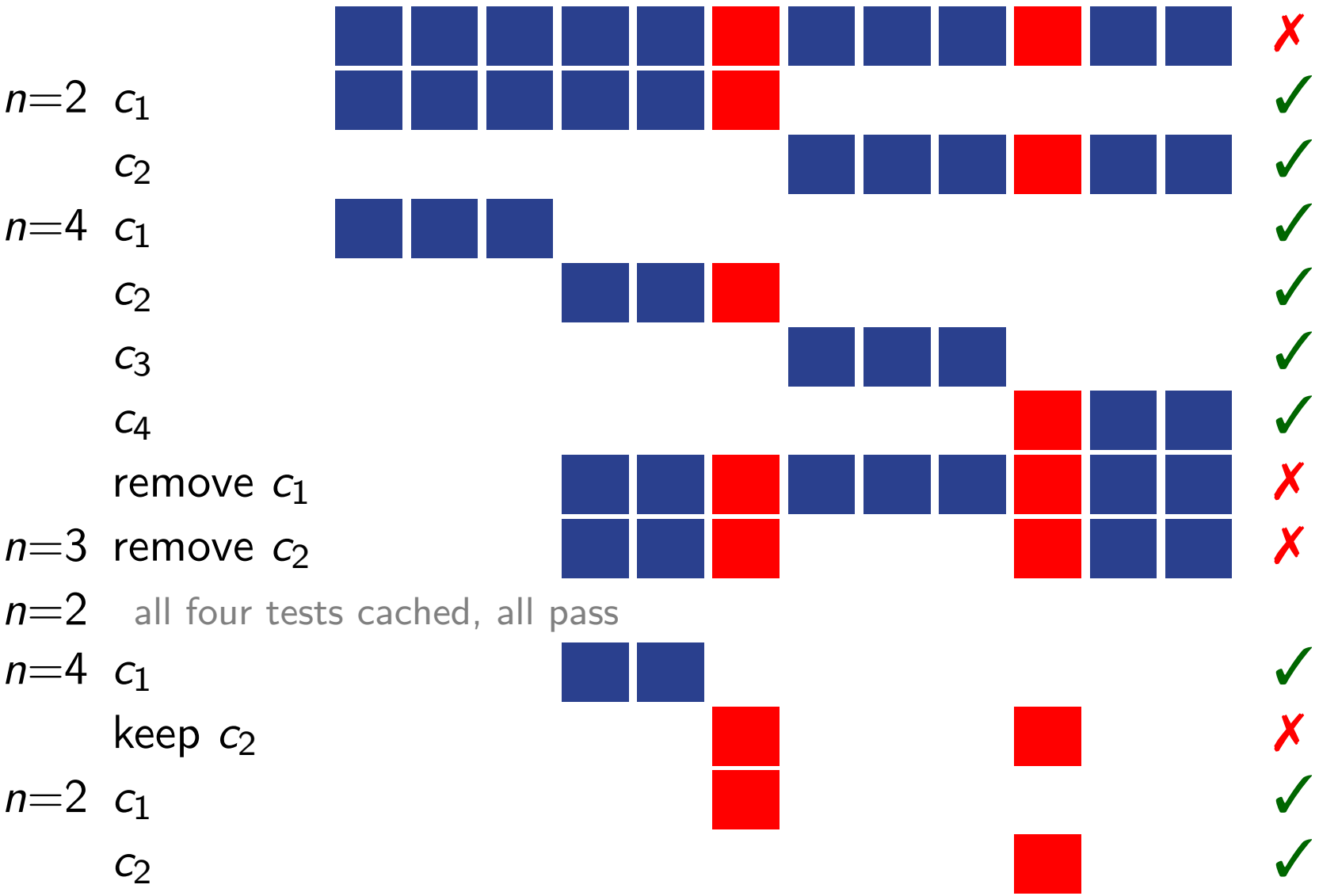


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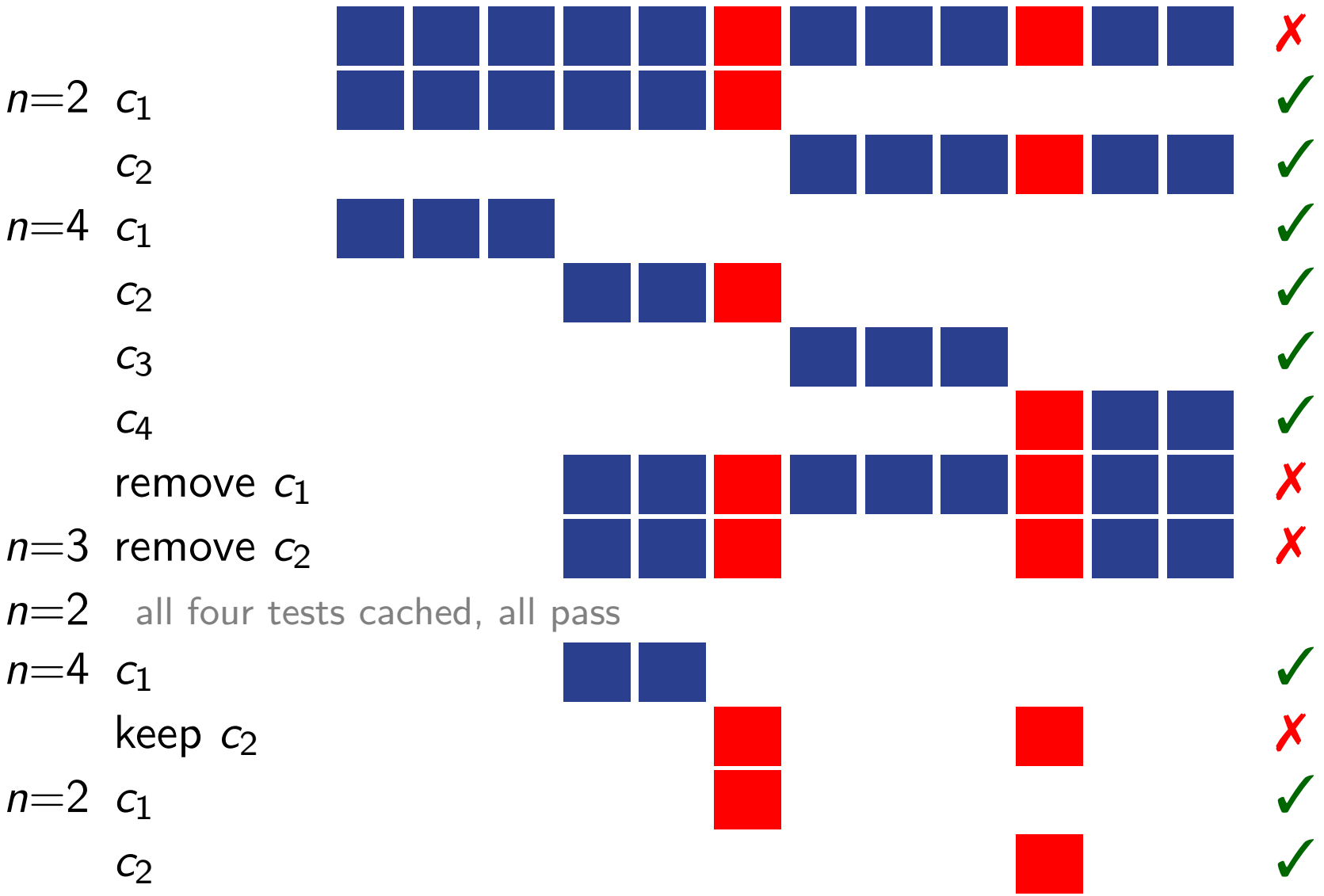
Recursive Delta Debugging – ddmin

PLRG



Recursive Delta Debugging – ddmin

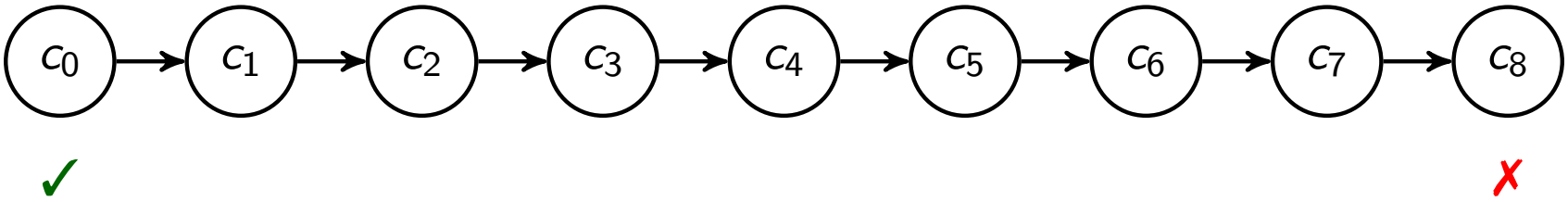
PLRG



12 runs, with repeated tests cached. Every rule fired.

Which Commit Broke It?

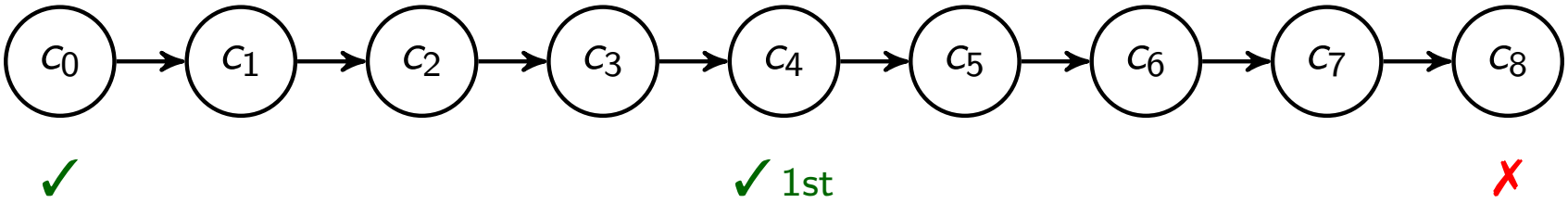
The parts can also be the **commits** between a passing and a failing version.[3]



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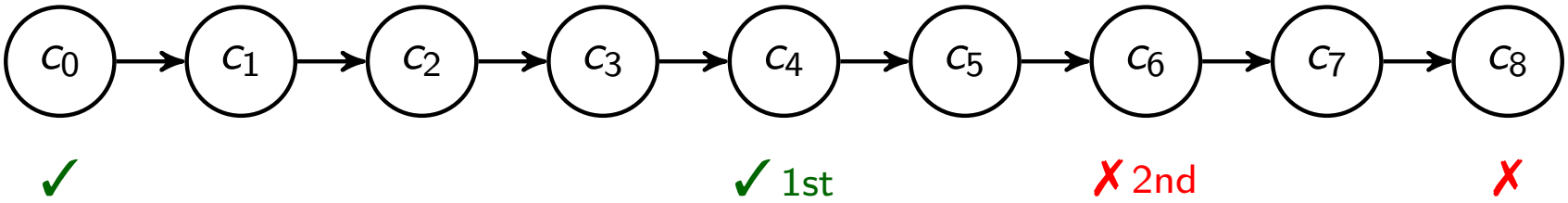
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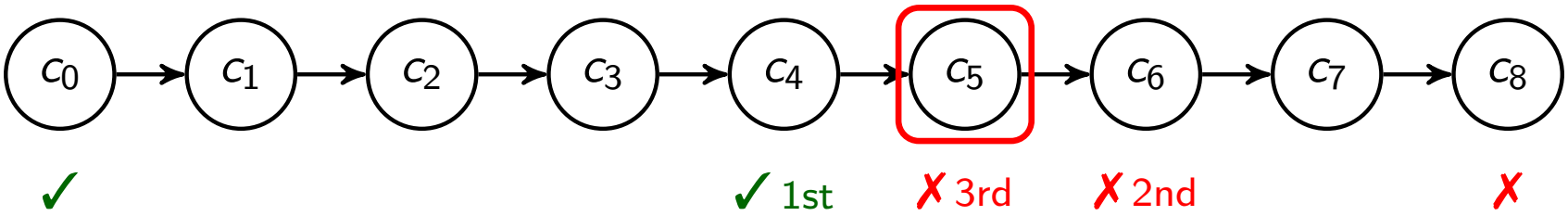
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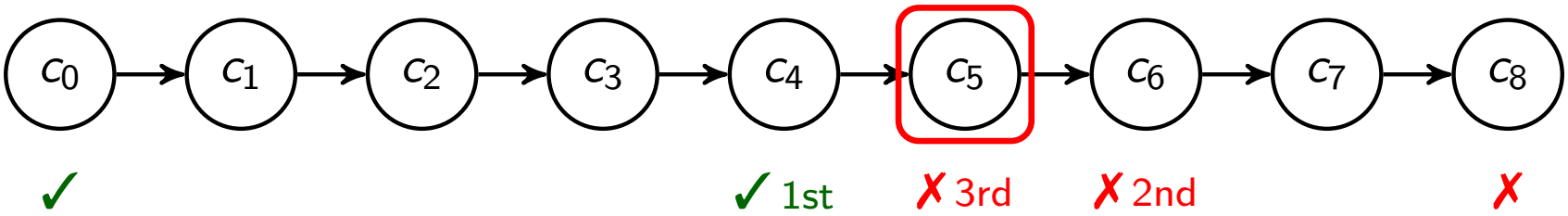


- One bad commit: **binary search**. c_5 in **3** runs out of 8. This is `git bisect`.

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- One bad commit: **binary search**. c_5 in **3** runs out of 8. This is `git bisect`.
- At Google, a **flaky** test can lie once and blame the wrong commit. So each run counts as noisy evidence, with a probability per commit.[4]

[3] **[ESEC/FSE'99]** A. Zeller. "*Yesterday, my Program Worked. Today, it Does Not. Why?*"

[4] **[ICST'23]** T. A. D. Henderson, B. Dorward, E. Nickell, C. Johnston, A. Kondareddy. "*Flake Aware Culprit Finding.*"

Hierarchical Delta Debugging (HDD)

Delete random characters from a **C program** and you get a **syntax error**, not the bug.[5]

- **HDD** runs `ddmin` on the **parse tree**, one level at a time: functions, then statements, then expressions.

[5] **[ICSE'06]** G. Misherghi, Z. Su. "*HDD: Hierarchical Delta Debugging.*"

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PLRG

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- **HDD** runs `ddmin` on the **parse tree**, one level at a time: functions, then statements, then expressions.
- Every try is a **valid** program.
- Compiler bugs are reported this way today (**C-Reduce**, **Perses**). New tools also ask an LLM for smaller rewrites.[6]

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Hierarchical Delta Debugging (HDD)

Level 1, **functions**: which of the three can go?

```
int copy(double to[],
         double from[], int count)
{
  int n= (count+7)/8;
  switch (count%8) do {
    case 0: *to++ = *from++;
    case 7: *to++ = *from++;
    case 6: *to++ = *from++;
    case 5: *to++ = *from++;
    case 4: *to++ = *from++;
    case 3: *to++ = *from++;
    case 2: *to++ = *from++;
    case 1: *to++ = *from++;
  } while (--n > 0);
  return (int)mult(to,2);
}
```

```
double mult( double z[], int n )
{
  int i;
  int j;
  for (j= 0; j< n; j++) {
    i= i+j+1;
    z[i]=z[i]*(z[0]+0);
  }
  return z[n];
}
int main( int argc, char *argv[] )
{
  double x[20], y[20];
  double *px= x;
  while (px < x + 20)
    *px++ = (px-x)*(20+1.0);
  return copy(y,x,20);
}
```

Hierarchical Delta Debugging (HDD)

Only `mult` survives. Level 2, **statements**:

```
double mult( double z[], int n )
{
  int i;
  int j;
  for (j= 0; j< n; j++) {
    i= i+j+1;
    z[i]=z[i]*(z[0]+0);
  }
  return z[n];
}
```

Hierarchical Delta Debugging (HDD)

Level 3, **expressions** inside what is left:

```
double mult( double z[], int n )
{
  int i;
  int j;
  for (j= 0; j< n; j++) {
    i= i+j+1;
    z[i]=z[i]*(z[0]+0);
  }
}
```

Hierarchical Delta Debugging (HDD)

PLRG

The final result. Every level removed only what the failure did not need:

```
double mult( double z[], int n )
{
  int i;
  int j;
  for (;;) {
    i= i+j+1;
    z[i]=z[i]*(z[0]+0);
  }
}
```

Probabilistic Delta Debugging (ProbDD)

`ddmin` cuts **blindly**. **ProbDD** keeps, for each element, the chance that the failure **needs** it, and deletes the least needed ones.[7]

	s1	s2	s3	s4	s5	s6	s7	s8	
	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	
1	s1	s2	s3	s4	s5	s6	s7	s8	F
	0.3657	0.3657	0.3657	0.2500	0.2500	0.2500	0.2500	0.3657	
2	s1	s2	s3	s4	s5	s6	s7	s8	T
	0.3657	0.3657	0.3657	0	0	0	0	0.3657	
3	s1	s2	s3	s4	s5	s6	s7	s8	F
	0.6119	0.3657	0.3657	0	0	0	0	0.6119	
4	s1	s2	s3	s4	s5	s6	s7	s8	F
	0.6119	0.6119	0.6119	0	0	0	0	0.6119	
5	s1	s2	s3	s4	s5	s6	s7	s8	T
	0.6119	0	0.6119	0	0	0	0	0.6119	
6	s1	s2	s3	s4	s5	s6	s7	s8	F
	0.6119	0	1	0	0	0	0	0.6119	
7	s1	s2	s3	s4	s5	s6	s7	s8	T
	0	0	1	0	0	0	0	0.6119	
8	s1	s2	s3	s4	s5	s6	s7	s8	F
	0	0	1	0	0	0	0	1	

Two rows per run: test the **teal** elements, then update every chance (**orange**). Still fails (**T**): the deleted ones become **0**. Passes (**F**): they go **up**.

[7] **[ESEC/FSE'21]** G. Wang, R. Shen, J. Chen, Y. Xiong, L. Zhang. "*Probabilistic Delta Debugging.*"

ProbDD Without the Probabilities

Look inside the math: each chance only goes **up**, one step per rejected deletion. It is just a **counter**.[8]

- **CDD**: one counter per element, delete the smallest first. Same results as ProbDD on 76 benchmarks.

[8] **[ICSE'25]** M. Zhang, Z. Xu, Y. Tian, X. Cheng, C. Sun. "*Toward a Better Understanding of Probabilistic Delta Debugging.*"

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- What really helped: **never repeat** a deletion already tried. The randomness did not matter.

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Look inside the math: each chance only goes **up**, one step per rejected deletion. It is just a **counter**.[8]

- **CDD**: one counter per element, delete the smallest first. Same results as ProbDD on 76 benchmarks.
- What really helped: **never repeat** a deletion already tried. The randomness did not matter.
- Lesson: when a technique works, ask **which part** works.

[8] **[ICSE'25]** M. Zhang, Z. Xu, Y. Tian, X. Cheng, C. Sun. "*Toward a Better Understanding of Probabilistic Delta Debugging.*"

Delta Debugging for Program Debloating

New test: not "*still crashes?*" but "*still does what I need?*" Then DD removes the rest.[9] Seven functions, `main` calls only `f1`. Plain DD: **16** runs.

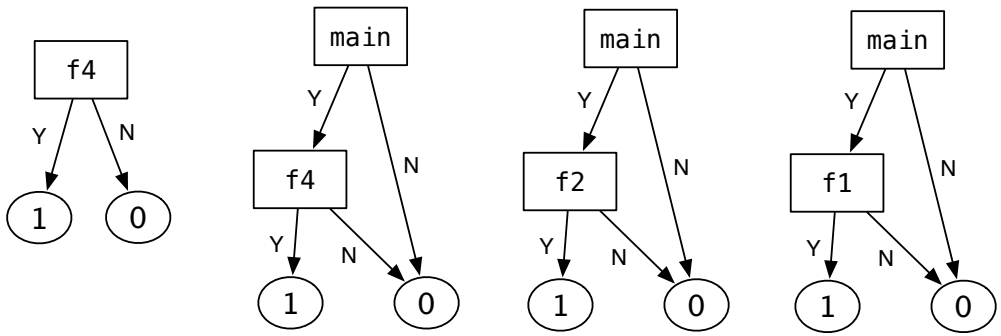
	f1	f2	f3	f4	f5	f6	f7	main	
1	f1	f2	f3	f4	f5	f6	f7	main	✘
2	f1	f2	f3	f4	f5	f6	f7	main	✘
3	f1	f2	f3	f4	f5	f6	f7	main	✘
4	f1	f2	f3	f4	f5	f6	f7	main	✘
5	f1	f2	f3	f4	f5	f6	f7	main	✘
6	f1	f2	f3	f4	f5	f6	f7	main	✘
7	f1	f2	f3	f4	f5	f6	f7	main	✘
8	f1	f2	f3	f4	f5	f6	f7	main	✔
9	f1	f2	f3	f4	f5	f6	f7	main	✔
10	f1	f2	f3	f4	f5	f6	f7	main	✘
11	f1	f2	f3	f4	f5	f6	f7	main	✘
12	f1	f2	f3	f4	f5	f6	f7	main	✘
13	f1	f2	f3	f4	f5	f6	f7	main	✘
14	f1	f2	f3	f4	f5	f6	f7	main	✘
15	f1	f2	f3	f4	f5	f6	f7	main	✔
16	f1	f2	f3	f4	f5	f6	f7	main	✔

[9] **[CCS'18]** K. Heo, W. Lee, P. Pashakhanloo, M. Naik. "*Effective Program Debloating via Reinforcement Learning.*"

Delta Debugging for Program Debloating

Chisel learns a decision tree from past runs to guess **which subset will pass**, and tries that subset first. A node asks whether a function is **kept**; a leaf guesses pass (**1**) or fail (**0**).

	f1	f2	f3	f4	f5	f6	f7	main	
1	f1	f2	f3	f4	f5	f6	f7	main	✘
2	f1	f2	f3	f4	f5	f6	f7	main	✘
3	f1	f2	f3	f4	f5	f6	f7	main	✘
4	f1	f2	f3	f4	f5	f6	f7	main	✔
5	f1	f2	f3	f4	f5	f6	f7	main	✔
6	f1	f2	f3	f4	f5	f6	f7	main	✘
7	f1	f2	f3	f4	f5	f6	f7	main	✔
8	f1	f2	f3	f4	f5	f6	f7	main	✔
9	f1	f2	f3	f4	f5	f6	f7	main	✘
10	f1	f2	f3	f4	f5	f6	f7	main	✘



Run 1 Run 2 Run 3 Run 6

- **Run 1**: two samples; any of `f1`–`f4` splits them. It picks `f4`.
- **Run 2**: `f1`–`f4` kept, still fails. `f4` is not enough; `main` goes on top.
- **Run 3**: `f3`–`main` kept, still fails. `f4` is out; `f2` takes over.
- **Run 6**: `f2`, `main` kept, still fails. `f2` → `f1`. Runs 4, 5, 7–10 fit.

Contents

1. Delta Debugging (DD)

2. Information Retrieval based Fault Localization (IRFL)
   - Vector Space Model (VSM)
   - Tf-Idf
   - Cleansing Bug Reports and Source Code

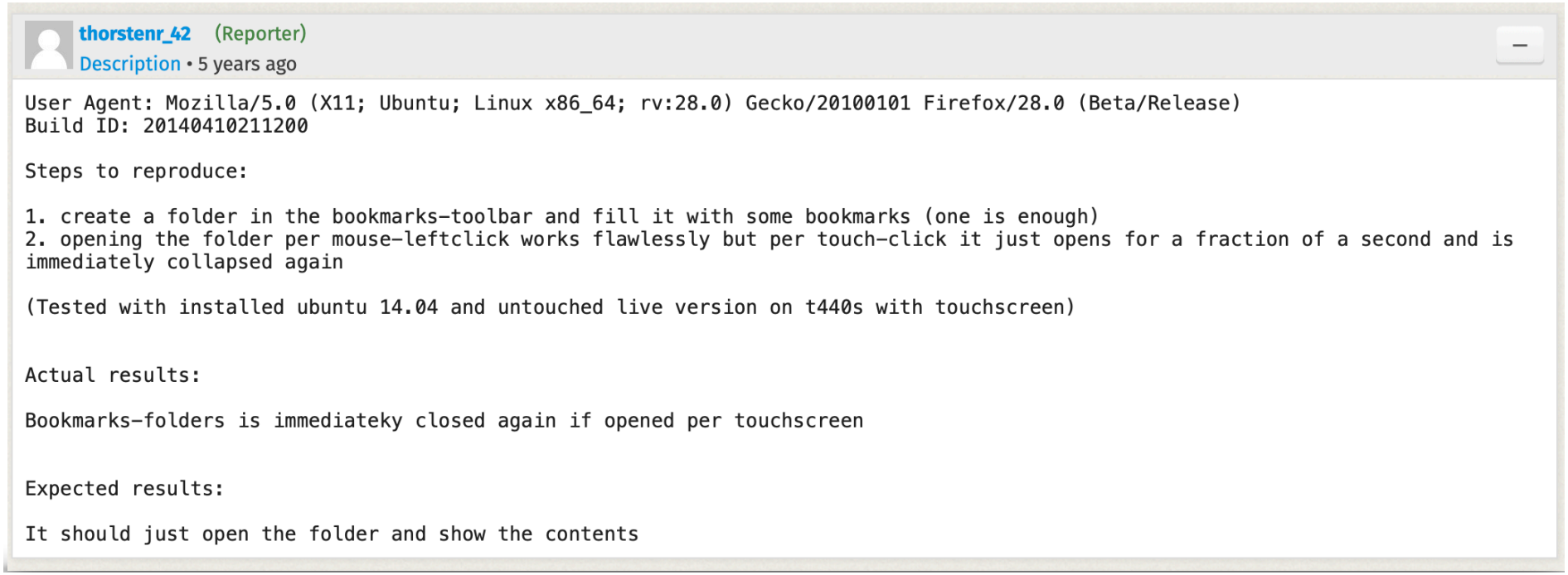
3. Spectrum-based Fault Localization (SBFL)

4. Mutation-based Fault Localization (MBFL)

5. Learning to Localize

Motivation

A bug report is **written in English** and names the things it is about. Can we use those words to find the file?[10]



thorstenr_42 (Reporter)
Description • 5 years ago

```
User Agent: Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:28.0) Gecko/20100101 Firefox/28.0 (Beta/Release)
Build ID: 20140410211200

Steps to reproduce:

1. create a folder in the bookmarks-toolbar and fill it with some bookmarks (one is enough)
2. opening the folder per mouse-leftclick works flawlessly but per touch-click it just opens for a fraction of a second and is
immediately collapsed again

(Tested with installed ubuntu 14.04 and untouched live version on t440s with touchscreen)


Actual results:

Bookmarks-folders is immediateky closed again if opened per touchscreen


Expected results:

It should just open the folder and show the contents
```

[10] **[Mozilla bug 1001903]** "*Can't open folder in Bookmarks toolbar with touchscreen: closes immediately.*"

Motivation

After **creating** *a* **folder** *in the* **bookmark toolbar** *and a* **bookmark** *inside the* **folder**, *mouse left-click successfully opens the folder but* **touch-click** *fails to* **open** *it: it* **opens** *but immediately* **closes** *itself.*

Motivation

PLRG

> *After* **creating** *a* **folder** *in the* **bookmark toolbar** *and a* **bookmark** *inside the* **folder**, *mouse left-click successfully opens the folder but* **touch-click** *fails to* **open** *it: it* **opens** *but immediately* **closes** *itself.*

You would open a file with **bookmark**, **folder**, and **touch** in it. **IRFL** does this: the report is the **query**, the files are the **documents**, and the answer is the most **similar** file.

Vector Space Model (VSM)

Count the words. The report and each file become a **vector**:

	folder	**bookmark**	**touch**	**open**	**close**	**print**
report q	2	2	1	2	1	0
`BookmarkFolder.js`	5	4	0	3	2	0
`TouchHandler.js`	0	0	6	2	1	0
`Printer.js`	0	0	0	1	0	7

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Cosine similarity: the cosine of the **angle** between the two vectors. For `BookmarkFolder.js` d and the report q:

$$\cos(d,q) = \frac{d \cdot q}{\|d\|\,\|q\|} = \frac{5{\cdot}2 + 4{\cdot}2 + 0{\cdot}1 + 3{\cdot}2 + 2{\cdot}1 + 0{\cdot}0}{\sqrt{5^2{+}4^2{+}0^2{+}3^2{+}2^2{+}0^2}\,\sqrt{2^2{+}2^2{+}1^2{+}2^2{+}1^2{+}0^2}}$$

$$= \frac{26}{\sqrt{54}\,\sqrt{14}} = 0.95$$

`BookmarkFolder.js`	`TouchHandler.js`	`Printer.js`
0.95	0.46	0.08

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`BookmarkFolder.js`	`TouchHandler.js`	`Printer.js`
0.95	0.46	0.08

The ranking **is** the answer: read `BookmarkFolder.js` first.

Tf-Idf

open is in every file, so it tells us nothing. Weight each word by how often it is in the file (**tf**) and how **rare** it is among the N files (**idf**):

$$tfidf(t, d) = \underbrace{f_{t,d}}_{\text{tf}} \times \underbrace{\log \frac{N}{n_t}}_{\text{idf}}$$

$f_{t,d}$: count of t in d

n_t: files with t

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	folder	**bookmark**	**touch**	**open**	**close**	**print**
$\log(3/n_t)$	1.10	1.10	1.10	**0**	0.41	1.10
report q	2.20	2.20	1.10	**0**	0.41	0
`BookmarkFolder.js`	5.49	4.39	0	**0**	0.81	0

folder in `BookmarkFolder.js`: $5 \times \log(3/1) = 5.49$.

open in `BookmarkFolder.js`: $3 \times \log(3/3) = 0$.

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open in `BookmarkFolder.js`: $3 \times \log(3/3) = 0$. New cosine: $d \cdot q \,/\, (\|d\| \, \|q\|) = 22.07 \,/\, (7.08 \cdot 3.33) = 0.94$.

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	`BookmarkFolder.js`	`TouchHandler.js`	`Printer.js`
counts	0.95	0.46	0.08
tf-idf	**0.94**	0.34	**0.00**

`Printer.js` matched the report only on **open**. Now it scores 0.

Cleansing Bug Reports and Source Code

Before counting, both sides are turned into the **same** kind of words:

Step	**Example**
Tokenize, lowercase	"Bookmarks toolbar" → `bookmarks`, `toolbar`
Stop words	"in the folder" → `folder`
Stemming	`closes`, `closing` → `close`
Split identifiers	`getFileName` → `get`, `file`, `name`
Keywords = stop words	`return`, `if`, `while` → dropped

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The last two are what makes **code** searchable by **English**: a report says "file name", the code says `getFileName`.

When the Report Names the File

Many reports already contain the answer:

```
Folder in bookmarks toolbar closes immediately on touch.
TypeError: this._popup is null
    at openFolder (BookmarkFolder.js:212)
    at onTouchEnd (TouchHandler.js:88)
```

- IRFL only has to **copy** the file name. Such reports make IRFL look better than it is.[11]

[11] **[ASE'14]** P. S. Kochhar, Y. Tian, D. Lo. "*Potential Biases in Bug Localization: Do They Matter?*"

[12] **[ICSE'12]** J. Zhou, H. Zhang, D. Lo. "*Where Should the Bugs Be Fixed? More Accurate Information Retrieval-Based Bug Localization Based on Bug Reports.*"

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- IRFL only has to **copy** the file name. Such reports make IRFL look better than it is.[11]
- **BugLocator** also uses the files fixed for **similar old reports**: the buggy file is in the top 10 (of 12,863) for 62.6% of Eclipse bugs.[12]

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   - Theoretical Analysis
   - Method-level Aggregation

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A Program and Its Tests

Back to `mid`. Six tests, one fails. Which tests **run** the faulty line?

```
1  int mid(int x, int y, int z) {
2    int m = z;
3    if (y < z) {
4      if (x < y)       m = y;
5      else if (x < z) m = y;   // bug
6    } else {
7      if (x > y)       m = y;
8      else if (x > z) m = x;
9    }
10   return m;
11 }
```

	input	**expected**	
t_1	$(3, 3, 5)$	3	✓
t_2	$(1, 2, 3)$	2	✓
t_3	$(3, 2, 1)$	2	✓
t_4	$(5, 5, 5)$	5	✓
t_5	$(5, 3, 4)$	4	✓
t_6	$(2, 1, 3)$	2	✗

A Program and Its Tests

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```

	input	expected	
t_1	$(3, 3, 5)$	3	✓
t_2	$(1, 2, 3)$	2	✓
t_3	$(3, 2, 1)$	2	✓
t_4	$(5, 5, 5)$	5	✓
t_5	$(5, 3, 4)$	4	✓
t_6	$(2, 1, 3)$	2	✗

Idea: a line run by the **failing** test but not by the **passing** ones is suspicious. Line 5 is run by t_6 – and by t_1.

Program Spectrum

PLRG

For each line, four numbers: covered by a **p**assing test (e_p), by a **f**ailing one (e_f), and **n**ot covered (n_p, n_f).

line	t_1	t_2	t_3	t_4	t_5	t_6	e_p	e_f	n_p	n_f
2 `m = z`	●	●	●	●	●	●	5	1	0	0
3 `if (y < z)`	●	●	●	●	●	●	5	1	0	0
4 `if (x < y)`	●	●			●	●	3	1	2	0
4 `m = y`		●					1	0	4	1
5 `else if (x < z)`	●				●	●	2	1	3	0
5 `m = y`	●					●	1	1	4	0
7 `if (x > y)`			●	●			2	0	3	1
7 `m = y`			●				1	0	4	1
8 `else if (x > z)`				●			1	0	4	1
8 `m = x`							0	0	5	1
10 `return m`	●	●	●	●	●	●	5	1	0	0

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line	t_1	t_2	t_3	t_4	t_5	t_6	e_p	e_f	n_p	n_f
2 `m = z`	●	●	●	●	●	●	5	1	0	0
3 `if (y < z)`	●	●	●	●	●	●	5	1	0	0
4 `if (x < y)`	●	●			●	●	3	1	2	0
4 `m = y`		●					1	0	4	1
5 `else if (x < z)`	●				●	●	2	1	3	0
5 `m = y`	●					●	1	1	4	0
7 `if (x > y)`			●	●			2	0	3	1
7 `m = y`			●				1	0	4	1
8 `else if (x > z)`				●			1	0	4	1
8 `m = x`							0	0	5	1
10 `return m`	●	●	●	●	●	●	5	1	0	0

The **spectrum** is all a formula sees: coverage, plus pass/fail per test.

Tarantula

The first SBFL formula, **Tarantula**[13]:

$$Tarantula(e) = \frac{\frac{e_f}{e_f+n_f}}{\frac{e_f}{e_f+n_f} + \frac{e_p}{e_p+n_p}}$$

[13] **[ICSE'02]** J. A. Jones, M. J. Harrold, J. Stasko. "*Visualization of Test Information to Assist Fault Localization.*" **[ASE'05]** J. A. Jones, M. J. Harrold. "*Empirical Evaluation of the Tarantula Automatic Fault-Localization Technique.*"

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$\underbrace{\frac{e_f}{e_f + n_f}}$ fraction of **failing** tests that cover e

$\underbrace{\frac{e_p}{e_p + n_p}}$ fraction of **passing** tests that cover e

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$$\underbrace{\frac{e_f}{e_f + n_f}}$$

fraction of **failing** tests that cover e

$$\underbrace{\frac{e_p}{e_p + n_p}}$$

fraction of **passing** tests that cover e

Suspicious = **run by failures, not by passes**. Every other formula is a variation of this.

[13] **[ICSE'02]** J. A. Jones, M. J. Harrold, J. Stasko. "*Visualization of Test Information to Assist Fault Localization.*" **[ASE'05]** J. A. Jones, M. J. Harrold. "*Empirical Evaluation of the Tarantula Automatic Fault-Localization Technique.*"

Tarantula on `mid`

One failing test, five passing. First fraction: 1 if t_6 runs the line. Second: $e_p/5$.

line	$\frac{e_f}{e_f+n_f}$	$\frac{e_p}{e_p+n_p}$	**Tarantula**	**rank**
5 `m = y`	1	0.2	**0.83**	**1**
5 `else if (x < z)`	1	0.4	0.71	2
4 `if (x < y)`	1	0.6	0.63	3
2, 3, 10	1	1.0	0.50	4
4 `m = y`, 7, 8	0	0–0.4	0	7

Tarantula on `mid`

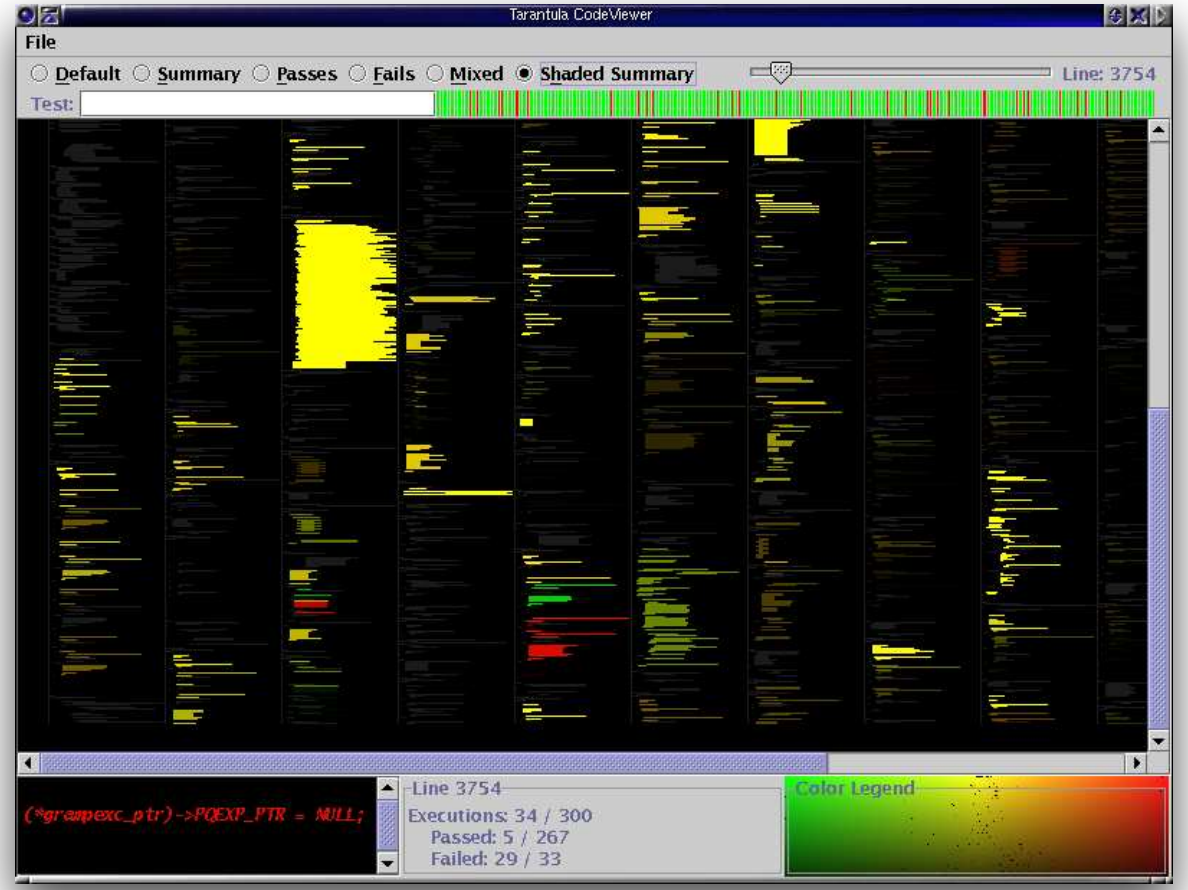
One failing test, five passing. First fraction: 1 if t_6 runs the line. Second: $e_p/5$.

line	$\frac{e_f}{e_f+n_f}$	$\frac{e_p}{e_p+n_p}$	**Tarantula**	**rank**
5 `m = y`	1	0.2	**0.83**	**1**
5 `else if (x < z)`	1	0.4	0.71	2
4 `if (x < y)`	1	0.6	0.63	3
2, 3, 10	1	1.0	0.50	4
4 `m = y`, 7, 8	0	0–0.4	0	7

The fault ranks **first**. Why only 0.83? t_1 also runs it and still passes: **coincidental correctness**.

Tarantula

It began as a **visualization**: color each line from red (only failures cover it) to green (only passes do).



Three Formulae, One Spectrum

Over 30 formulae exist. Two common ones – **Ochiai**, the usual default,[14] and **Op2**:[15]

$$Ochiai = \frac{e_f}{\sqrt{(e_f + n_f)(e_f + e_p)}} \qquad Op2 = e_f - \frac{e_p}{e_p + n_p + 1}$$

	`5 m = y`	`5 else if`	`4 if`	**2, 3, 10**
(e_f, e_p)	$(1,1)$	$(1,2)$	$(1,3)$	$(1,5)$
Tarantula	**0.83**	0.71	0.63	0.50
Ochiai	**0.71**	0.58	0.50	0.41
Op2	**0.83**	0.67	0.50	0.17

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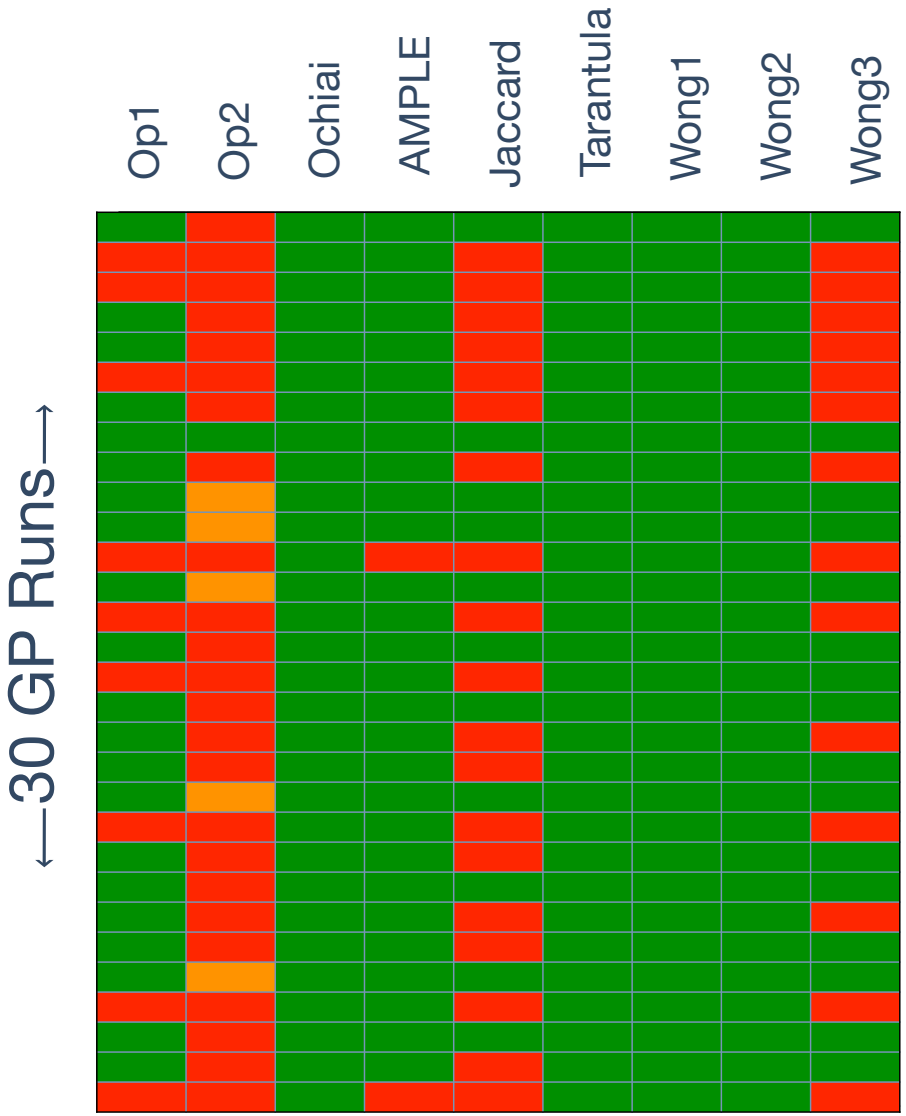
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Different numbers, **same order**. With one failing test, each formula just sorts by e_p.

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Genetic Programming for SBFL

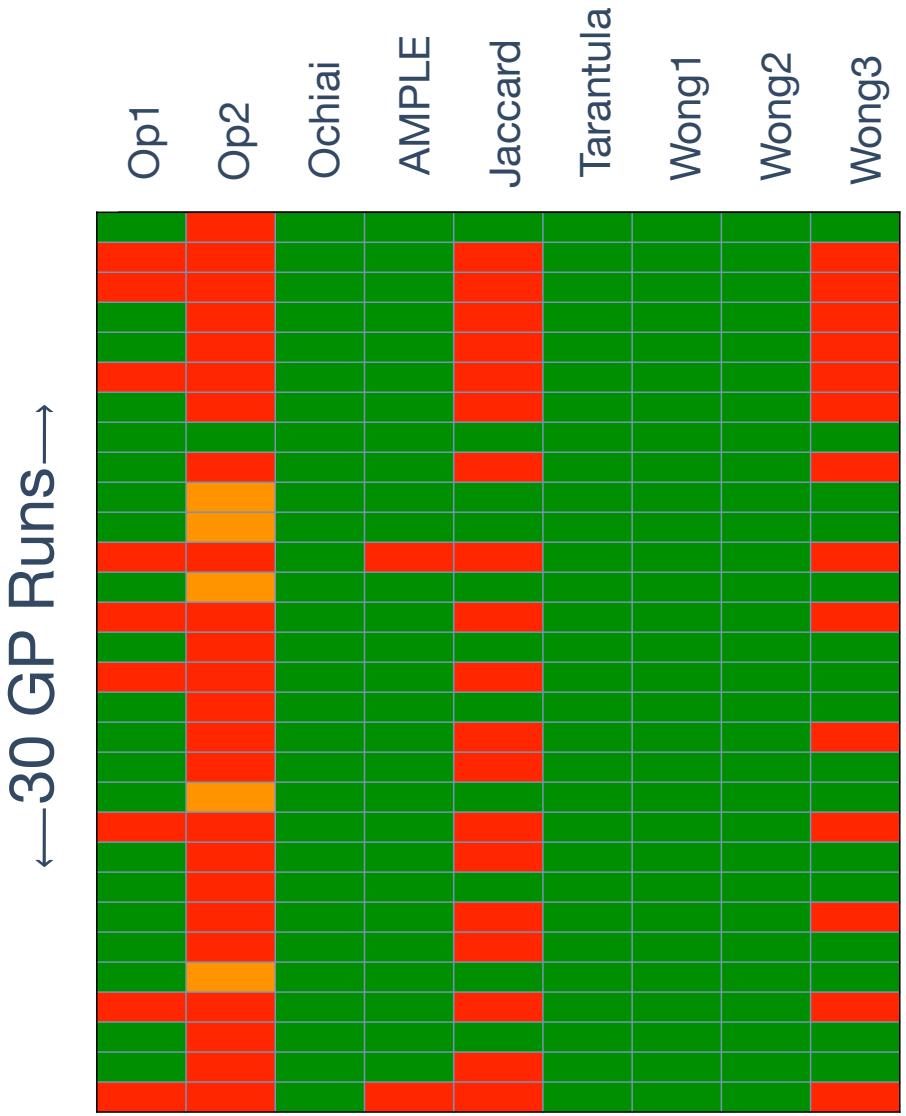


4 Unix tools w/ 92 faults: 20 for training, 72 for evaluation.

- A formula is a tree over e_f, e_p, n_f, n_p. So **evolve** it. Fitness: the rank of the real fault.

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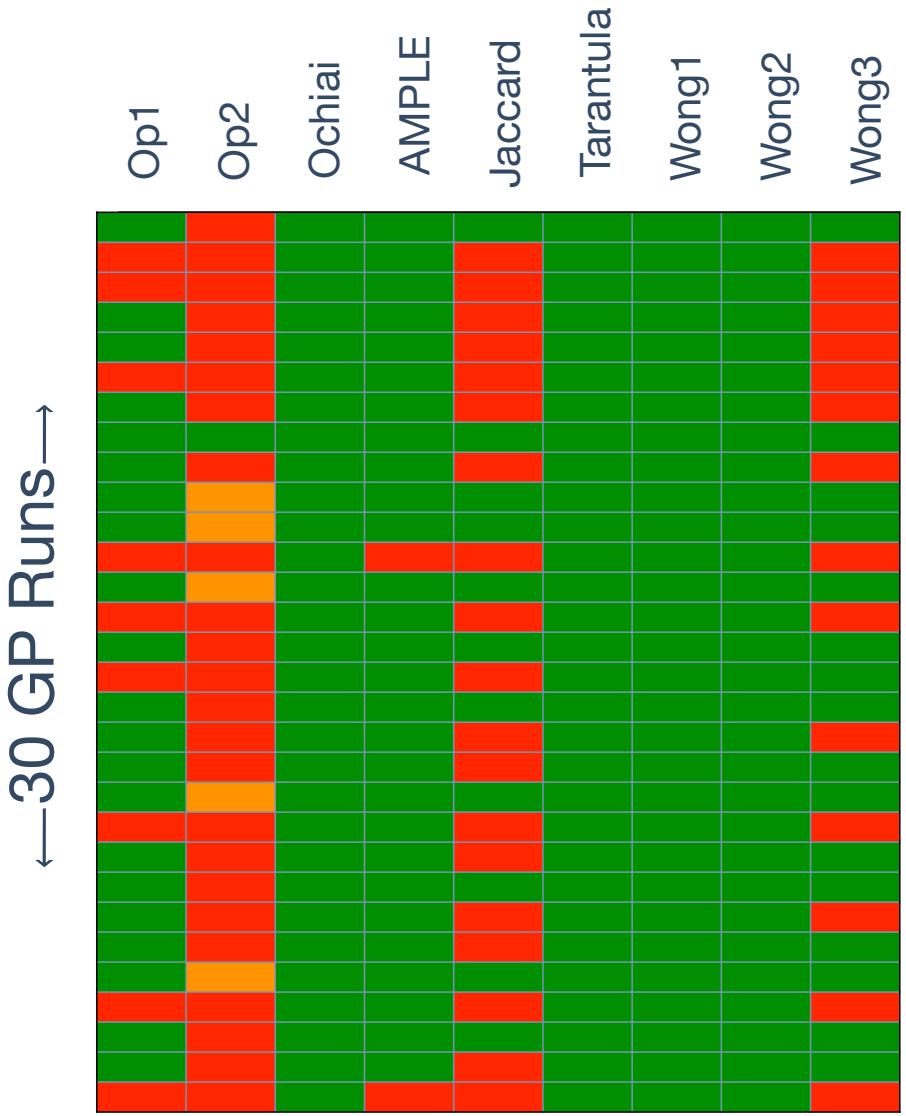


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- 30 GP runs vs. 9 human formulae. Green: GP wins, orange: tie, red: human wins.
- GP **always** beats Tarantula and Ochiai. It almost **never** beats Op2.

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Is There a Best Formula?

Two failing tests f_1, f_2 and four passing. Line A is run by f_1 only. Line B is run by both failures and two passes.

	(e_f, n_f, e_p, n_p)	**Tarantula**	**Ochiai**	**Op2**
A	$(1, 1, 0, 4)$	**1.00**	0.71	1.0
B	$(2, 0, 2, 2)$	0.67	0.71	**1.6**

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- Op2 $= e_f - \frac{e_p}{e_p+n_p+1}$: the pass term is always below 1, so **more failing runs always wins**. A line run by every failing test is always on top.
- Proven: with one fault, no formula can rank better than this.[16] So GP could only **tie** Op2, not beat it.

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Method-level Aggregation

```
public void testMe(int a) {
  util();
  if (a % 3 == 0) {
    ... // faulty code
  } else {
    ...
  }
}
public void util() {
  ...
}
```

- Every test that reaches the fault also runs `util()`. So both methods get the **same** score.

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- Every test that reaches the fault also runs `util()`. So both methods get the **same** score.
- Fix: rank **statements**, then give each method the score of its **best** statement. `testMe` wins.

Developers read **methods**, so rank methods.[17]

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What a Spectrum Cannot See

```
int div(int a, int b) {
  // missing: if (b == 0) ...
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Missing code: the faulty line does not **exist**, so no test runs it. SBFL points at `return a / b`, the symptom.

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- **Many faults**: their failing tests mix in one spectrum, and each fault lowers the other's score.
- **Coincidental correctness**: like t_1 in `mid`, each test that runs the bug and passes pushes the fault **down**.

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Mutation-based Fault Localization (MBFL)

A spectrum only says a line **ran**. MBFL **changes** the line (a **mutant**) and runs the tests again.[18]

change a ...	**failing tests**	**passing tests**
faulty line	some now **pass** (fixed)	few now fail
correct line	still fail	many now **fail** (broken)

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faulty line	some now **pass** (fixed)	few now fail
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So a line is **more suspicious** when changing it **fixes** failing tests, and **less** when changing it **breaks** passing ones.

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MUSE

MUSE: **add** the failing tests a mutant fixes, **subtract** the passing tests it breaks. Average over the mutants $mut(s)$ of s:[19]

$$\mu(s) = \frac{1}{|mut(s)|} \sum_{m \in mut(s)} \left(\underbrace{\frac{|f_P(s) \cap p_m|}{|f_P|}}_{\text{fail}\rightarrow\textbf{pass}} - \alpha \cdot \underbrace{\frac{|p_P(s) \cap f_m|}{|p_P|}}_{\text{pass}\rightarrow\textbf{fail}} \right)$$

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- α: pass $\to$ fail is much more common, so it gets a smaller weight.

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MUSE on `mid`

35 mutants: change each comparison operator, and each assigned variable. Run all 6 tests on each and count the tests that **flip**:

statement	**mutants**	**fail → pass**	**pass → fail**	$\mu(s)$
line 5 `m = y`	2	**1**	1	**0.48**
line 8	7	0	0	0.00
all the others	26	0	30	-0.02 to -0.08

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- The only fail → pass: `m = y` → `m = x` on line 5. That mutant **is the fix**.
- A clearer gap than Tarantula (0.48 vs. 0, not 0.83 vs. 0.71). But it costs 35 mutants × 6 tests = **210** runs. A spectrum needs **6**.

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    Fault Localization in Practice

Learning to Rank

No formula wins everywhere. So give **many** signals to a ranker **learned** from known faults:

Approach	**Features**
Multric[20]	25 formulae, with learned weights
Savant[21]	35 formulae + **invariants** that differ between passing and failing runs
FLUCCS[22]	formulae + code **age**, **changes**, **size**

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PLRG

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Fault localization became a **learning** problem.

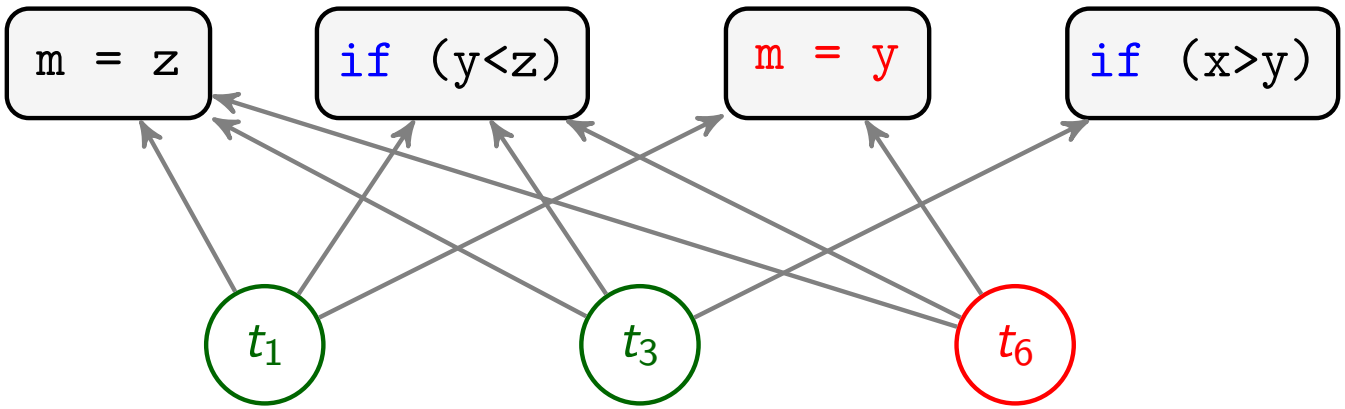
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Learning From the Spectrum Itself

Why pick formulae by hand? **GRACE** gives the whole coverage to a graph neural network.[23]

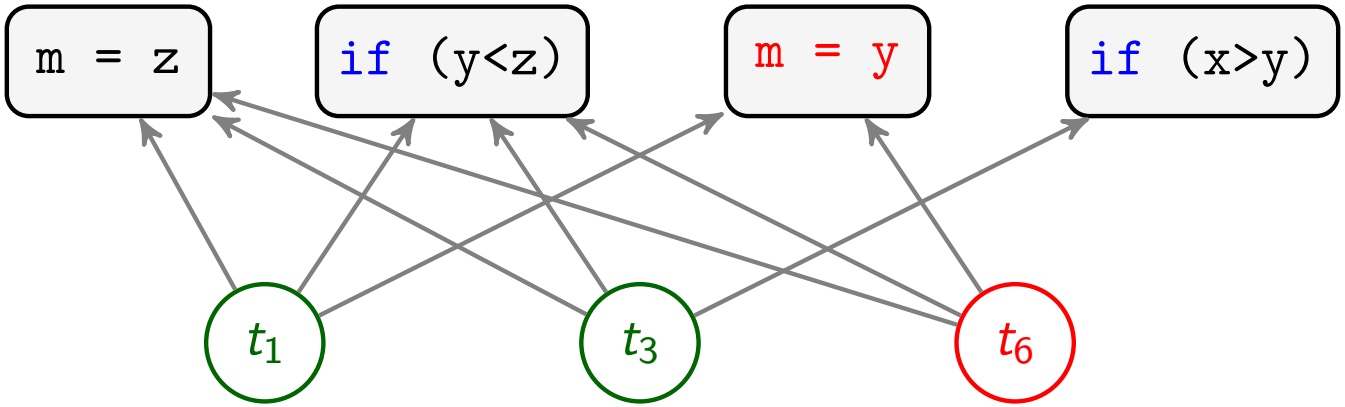


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[23] **[ESEC/FSE'21]** Y. Lou, Q. Zhu, J. Dong, X. Li, Z. Sun, D. Hao, L. Zhang, L. Zhang. "*Boosting Coverage-Based Fault Localization via Graph-Based Representation Learning.*"

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- Nodes: statements and tests. Edges: coverage. Nothing is squeezed into four numbers.
- Defects4J, top-1: **195** bugs, vs. **166** for the best older method.

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Ask a Language Model

AutoFL: show an LLM the failing test. It calls **tools** to read the code, then names the buggy method and says **why**. Ask **several times**: how often it names a method is its score.[24]

```
Test: mid(2, 1, 3) expected 2, got 1
LLM > get_covered_methods()  -> [mid]
LLM > get_code_snippet(mid)  -> "int mid(int x, ..."
LLM : "1 < 2 < 3, so the answer is x = 2.
       That case reaches line 5, which returns y."
      Culprit: mid
```

[24] **[FSE'24]** S. Kang, G. An, S. Yoo. "*A Quantitative and Qualitative Evaluation of LLM-Based Explainable Fault Localization.*"

Fault Localization in Practice

- **Defects4J**: hundreds of real Java bugs, each with a failing test and the real fix.[25]

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PLRG

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- **acc@*n***: the number of bugs whose faulty element is in the top *n*.
- **No family wins alone.** SBFL, MBFL, slicing, and IRFL find different bugs. Together they beat each one.[26] DeepFL learns how to combine them.[27]

---

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- The next lecture has a reader that will read **hundreds**: an **automatic repair** tool.

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Homework #4 – Fault Localization

PLRG

- Given a **faulty program**, an **oracle** (the correct program, callable but not readable), and one **failing input**, rank the statements by how likely each is to contain the fault.

- Graded mostly on **hidden** programs: they are larger and harder than the public examples, so a perfect score on the examples does not mean a perfect grade.

- Please see this document on GitHub:

  https://github.com/ku-plrg-classroom/js-fl

- The due date is **23:59 on October 26 (Mon.)**, and please submit it to **LMS**.

Next Lecture

- Automated Program Repair

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